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Methods and systems for managing access to implantable medical devices

Abstract

An external device for wireless communication with an implantable medical device can include a memory having stored thereon instructions for a first application and authentication credentials specific to the first application; and a processor configured for transmitting the authentication credentials specific to the first application to an implantable medical device to authorize access to the implantable medical device by the first application. An external device for wireless communication with an implantable medical device can include a memory including authentication credentials stored thereon; and a processor configured for transmitting the authentication credentials to an implantable medical device to authorize access; receiving a request to facilitate authorization of the second device and receiving second authentication credentials from the second device; determining authentication or authorization of the second device; and transmitting a token or authorization credentials directing the implantable medical device to authorize access by the second device.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Patent Application Ser. No. 63/109,749, filed Nov. 4, 2020, which is incorporated herein by reference.

FIELD

(1) The present disclosure is directed to the area of methods and systems for managing access to implantable medical devices from external devices. The present disclosure is also directed to methods and systems for managing access to implantable medical devices using authentication credentials.

BACKGROUND

(2) Implantable electrical stimulation systems have proven therapeutic in a variety of diseases and disorders. For example, deep brain stimulation systems have been used as a therapeutic modality for the treatment of Parkinson's disease, essential tremor, and the like.

(3) Stimulators have been developed to provide therapy for a variety of treatments. A stimulator can include an implantable pulse generator (IPG), one or more leads, and an array of stimulator electrodes on each lead. The stimulator electrodes are in contact with or near the nerves, muscles, or other tissue to be stimulated. The pulse generator in the IPG generates electrical pulses that are delivered by the electrodes to body tissue.

(4) Implantable medical devices (IMDs), including IPGs, typically have the capability to communicate data with an external device, such as a clinician programmer or a remote control, via a radio-frequency telemetry link or other wireless communication method. The clinician programmer can program the operating parameters of the implanted medical device. The remote control can switch programs. Modern implantable devices also include the capability for bidirectional communication so that information can be transmitted to the clinician programmer or remote control from the implanted device.

BRIEF SUMMARY

(5) One aspect is an external device for wireless communication with an implantable medical device. The external device includes a memory having stored thereon instructions for a first application configured to interact with implantable medical devices and authentication credentials specific to the first application; and a processor coupled to the memory and configured to perform actions including: executing the first application using the instructions; and transmitting the authentication credentials specific to the first application to an implantable medical device for the implantable medical device to authorize access to the implantable medical device by the first application.

(6) In at least some aspects, the memory further has stored thereon instructions for a second application configured to interact with implantable medical devices and authentication credentials specific to the second application. In at least some aspects, the authentication credentials specific to the first application include identification of at least one specific implantable medical device or at least one specific type of implantable medical device that the first application is configured to interact with.

- (7) In at least some aspects, the authentication credentials specific to the first application include an end-entity certificate. In at least some aspects, the end-entity certificate includes an identification of the first application in custom metadata or certificate extensions of the end-entity certificate.
- (8) In at least some aspects, the authentication credentials specific to the first application include identification of at least one interaction capability of the first application with respect to an implantable medical device. In at least some aspects, at least one of the at least one interaction capability is selected from read data, write data, select from existing programs, modify programming settings of existing programs, create programs, modify or update software/firmware, or full control. In at least some aspects, the authentication credentials specific to the first application include an end-entity certificate and the end-entity certificate includes the identification of the at least one interaction capability in custom metadata or certificate extensions of the end-entity certificate.
- (9) Another aspect is a system that includes any of the external devices described above and an implantable medical device configured for wireless communication with the external device.
- (10) In at least some aspects, the authentication credentials specific to the first application include identification of the implantable medical device in the authentication credentials. In at least some aspects, the authentication credentials specific to the first application include identification of at least one interaction capability of the first application with respect to an implantable medical device and the implantable medical device is configured to limit authorized access to the implantable medical device by the first application to only the at least one interaction capability identified in the authentication credentials.
- (11) Yet another aspect is an external device for wireless communication with an implantable medical device. The external device includes a memory including authentication credentials stored thereon; and a processor coupled to the memory and configured to perform actions including: transmitting the authentication credentials to an implantable medical device for the implantable medical device to authorize access to the implantable medical device by the external device; receiving a request to facilitate authorization of the second device by the implantable medical device and receiving second authentication credentials from the second device; determining authentication or authorization of the second device using the second authentication credentials; and after authorization of the external device by the implantable medical device and determining authentication or authorization of the second device, transmitting a token or authorization credentials to at least one of the implantable medical device or the second device directing the implantable medical device to authorize access to the implantable medical device by the second device.
- (12) In at least some aspects, the token or authorization credentials are limited to a specified time period. In at least some aspects, the token or authorization credentials are limited to a single session.
- (13) A further aspect is a system that includes any of the external devices described in the preceding two paragraphs and a second device including a memory having stored thereon instructions for a second application configured to interact with implantable medical devices and the second authentication credentials, and a processor coupled to the memory and configured to perform actions including: executing the second application using the instructions; and transmitting the second authentication credentials specific to the external device.
- (14) In at least some aspects, the memory of the external device has further stored thereon instructions for a first application configured to interact with implantable medical devices, wherein the authentication credentials are specific to the first application and identify of at least one interaction capability of the first application with respect to an implantable medical device, and wherein the second authentication credentials identify of at least one interaction capability of the second application with respect to an implantable medical device.
- (15) In at least some aspects, the token or authorization credentials identify the at least one

interaction capability of the first application. In at least some aspects, the token or authorization credentials identify the at least one interaction capability of the second application.

(16) In at least some aspects, the system further includes the implantable medical device configured for wireless communication with the external device and the second device. In at least some aspects, the implantable medical device is configured to limit authorized access to the second device based on identification of at least one interaction capability in the token or authorization credentials.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Non-limiting and non-exhaustive embodiments of the present invention are described with reference to the following drawings. In the drawings, like reference numerals refer to like parts throughout the various figures unless otherwise specified.
- (2) For a better understanding of the present invention, reference will be made to the following Detailed Description, which is to be read in association with the accompanying drawings, wherein:
- (3) FIG. 1 is a schematic view of one embodiment of an electrical stimulation system that includes one or more leads that can be coupled to an IPG;
- (4) FIG. 2 is a block diagram of element of an electrical stimulation system;
- (5) FIG. 3 is a schematic illustration of authentication between an IPG and two external devices;
- (6) FIG. 4 is a schematic illustration of authentication between an IPG and two applications on an external device;
- (7) FIG. 5 is a flowchart of one embodiment of method of authorizing access to an IPG by an external device;
- (8) FIG. 6 is a schematic illustration of authentication between an IPG and external devices or applications of an external device and also indicating interaction capabilities of the external devices or applications;
- (9) FIG. 7 is a flowchart of one embodiment of method of authorizing access to an IPG by an external device and authorizing interaction capabilities by the external device or an application on the external device;
- (10) FIG. 8 is a schematic illustration of access authorization to an IPG by a second external device using an authorized first external device to authenticate the second external device; and
- (11) FIG. 9 is a flowchart of one embodiment of method of authorizing access to an IPG by a second external device using an authorized first external device to authenticate the second external device.

DETAILED DESCRIPTION

- (12) The present disclosure is directed to the area of methods and systems for managing access to implantable medical devices from external devices. The present disclosure is also directed to methods and systems for managing access to implantable medical devices using authentication credentials.
- (13) The systems and methods described herein can be used with any suitable implantable medical devices. Implantable electrical stimulation systems and devices are used herein to exemplify the inventions, but it will be understood that these inventions can be utilized with other implantable medical devices. Examples of implantable electrical stimulation systems include, but are not limited to, a least one lead with one or more electrodes disposed along a distal end of the lead and one or more terminals disposed along the one or more proximal ends of the lead. Examples of electrical stimulation systems with leads are found in, for example, U.S. Pat. Nos. 6,181,969; 6,295,944; 6,391,985; 6,516,227; 6,609,029; 6,609,032; 6,741,892; 7,244,150; 7,450,997; 7,672,734; 7,761,165; 7,783,359; 7,792,590; 7,809,446; 7,949,395; 7,974,706; 8,831,742;

8,688,235; 8,175,710; 8,224,450; 8,271,094; 8,295,944; 8,364,278; and 8,391,985; U.S. Patent Application Publications Nos. 2007/0150036; 2009/0187222; 2009/0276021; 2010/0076535; 2010/0268298; 2011/0004267; 2011/0078900; 2011/0130817; 2011/0130818; 2011/0238129; 2011/0313500; 2012/0016378; 2012/0046710; 2012/0071949; 2012/0165911; 2012/0197375; 2012/0203316; 2012/0203320; 2012/0203321; 2012/0316615; 2013/0105071; 2011/0005069; 2010/0268298; 2011/0130817; 2011/0130818; 2011/0078900; 2011/0238129; 2011/0313500; 2012/0016378; 2012/0046710; 2012/0165911; 2012/0197375; 2012/0203316; 2012/0203320; and 2012/0203321, all of which are incorporated by reference in their entireties.

(14) Turning to FIG. 1, one embodiment of an electrical stimulation system **10** includes one or more stimulation leads **12** and an implantable pulse generator (IPG) **14**. The system **10** can also include one or more of an external remote control (RC) **16**, a clinician's programmer (CP) **18**, an external trial stimulator (ETS) **20**, or an external charger **22**. The IPG and ETS are examples of control modules for the electrical stimulation system.

(15) The IPG **14** is physically connected, optionally via one or more lead extensions **24**, to the stimulation lead(s) **12**. Each lead carries multiple electrodes **26** arranged in an array. The IPG **14** includes pulse generation circuitry that delivers electrical stimulation energy in the form of, for example, a pulsed electrical waveform (i.e., a temporal series of electrical pulses) to the electrode array **26** in accordance with a set of stimulation parameters. The implantable pulse generator can be implanted into a patient's body, for example, below the patient's clavicle area or within the patient's abdominal cavity or at any other suitable site. The implantable pulse generator **14** can have multiple stimulation channels which may be independently programmable to control the magnitude of the current stimulus from each channel. In some embodiments, the implantable pulse generator **14** can have any suitable number of stimulation channels including, but not limited to, 4, 6, 8, 12, 16, 32, or more stimulation channels. The implantable pulse generator **14** can have one, two, three, four, or more connector ports, for receiving the terminals of the leads and/or lead extensions.

(16) The ETS **20** may also be physically connected, optionally via the percutaneous lead extensions **28** and external cable **30**, to the stimulation leads **12**. The ETS **20**, which may have similar pulse generation circuitry as the IPG **14**, also delivers electrical stimulation energy in the form of, for example, a pulsed electrical waveform to the electrode array **26** in accordance with a set of stimulation parameters. One difference between the ETS **20** and the IPG **14** is that the ETS **20** is often a non-implantable device that is used on a trial basis after the neurostimulation leads **12** have been implanted and prior to implantation of the IPG **14**, to test the responsiveness of the stimulation that is to be provided. Any functions described herein with respect to the IPG **14** can likewise be performed with respect to the ETS **20**.

(17) The RC **16** may be used to telemetrically communicate with or control the IPG **14** or ETS **20** via a uni- or bi-directional wireless communications link **32**. Once the IPG **14** and neurostimulation leads **12** are implanted, the RC **16** may be used to telemetrically communicate with or control the IPG **14** via a uni- or bi-directional communications link **34**. Such communication or control allows the IPG **14**, for example, to be turned on or off and to be programmed with different stimulation parameter sets. The IPG **14** may also be operated to modify the programmed stimulation parameters to actively control the characteristics of the electrical stimulation energy output by the IPG **14**. In at least some embodiments, the CP **18** allows a user, such as a clinician, the ability to program stimulation parameters for the IPG **14** and ETS **20** in the operating room and in follow-up sessions. Alternately, or additionally, in at least some embodiments, stimulation parameters can be programed via wireless communications (e.g., Bluetooth) between the RC **16** (or external device such as a hand-held electronic device like a mobile phone, tablet, or the like) and the IPG **14**.

(18) The CP **18** may perform this function by indirectly communicating with the IPG **14** or ETS **20**, through the RC **16**, via a wireless communications link **36**. Alternatively, the CP **18** may directly communicate with the IPG **14** or ETS **20** via a wireless communications link (not shown). The stimulation parameters provided by the CP **18** are also used to program the RC **16**, so that the

stimulation parameters can be subsequently modified by operation of the RC **16** in a stand-alone mode (i.e., without the assistance of the CP **18**).

(19) Additional examples of the RC **16**, CP **18**, ETS **20**, and external charger **22** can be found in the references cited herein as well as U.S. Pat. Nos. 6,895,280; 6,181,969; 6,516,227; 6,609,029; 6,609,032; 6,741,892; 7,949,395; 7,244,150; 7,672,734; and 7,761,165; 7,974,706; 8,175,710; 8,224,450; and 8,364,278; and U.S. Patent Application Publication No. 2007/0150036, all of which are incorporated herein by reference in their entireties.

(20) FIG. **2** is a schematic overview of one embodiment of components of an electrical stimulation system **200** including an electronic subassembly **210** disposed within an IPG **14** (FIG. **1**). It will be understood that the electrical stimulation system can include more, fewer, or different components and can have a variety of different configurations including those configurations disclosed in the stimulator references cited herein.

(21) The IPG **14** (FIG. **1**) can include, for example, a power source **212**, antenna **218**, receiver **202**, processor **204**, and memory **205**. An external device, such as a CP or RC **206**, can include a processor **207** and memory **208**. Some of the components (for example, power source **212**, antenna **218**, receiver **202**, processor **204**, and memory **205**) of the electrical stimulation system can be positioned on one or more circuit boards or similar carriers within a sealed housing of the IPG **14** (FIG. **1**), if desired. Any power source **212** can be used including, for example, a battery such as a primary battery or a rechargeable battery. Examples of other power sources include super capacitors, nuclear or atomic batteries, mechanical resonators, infrared collectors, thermally-powered energy sources, flexural powered energy sources, bioenergy power sources, fuel cells, bioelectric cells, osmotic pressure pumps, and the like including the power sources described in U.S. Pat. No. 7,437,193, incorporated herein by reference in its entirety.

(22) As another alternative, power can be supplied by an external power source through inductive coupling via the optional antenna **218** or a secondary antenna. The external power source can be in a device that is mounted on the skin of the user or in a unit that is provided near the user on a permanent or periodic basis.

(23) If the power source **212** is a rechargeable battery, the battery may be recharged using the optional antenna **218**, if desired. Power can be provided to the battery for recharging by inductively coupling the battery through the antenna to a recharging unit **216** external to the user. Examples of such arrangements can be found in the references identified above.

(24) In one embodiment, electrical current is emitted by the electrodes **26** on the lead body to stimulate nerve fibers, muscle fibers, or other body tissues near the electrical stimulation system. A processor **204** is generally included to control the timing and electrical characteristics of the electrical stimulation system. For example, the processor **204** can, if desired, control one or more of the timing, frequency, amplitude, width, and waveform of the pulses. In addition, the processor **204** can select which electrodes can be used to provide stimulation, if desired. In some embodiments, the processor **204** may select which electrode(s) are cathodes and which electrode(s) are anodes. In some embodiments, the processor **204** may be used to identify which electrodes provide the most useful stimulation of the desired tissue. Instructions for the processor **204** can be stored on the memory **205**. Instructions for the processor **207** can be stored on the memory **208**.

(25) Any processor **204** can be used for the IPG and can be as simple as an electronic device that, for example, produces pulses at a regular interval or the processor can be capable of receiving and interpreting instructions from the CP/RC **206** (such as CP **18** or RC **16** of FIG. **1**) that, for example, allows modification of pulse characteristics. In the illustrated embodiment, the processor **204** is coupled to a receiver **202** which, in turn, is coupled to the optional antenna **218**. This allows the processor **204** to receive instructions from an external source to, for example, direct the pulse characteristics and the selection of electrodes, if desired. Any suitable processor **207** can be used for the CP/RC **206**.

(26) Any suitable memory **205**, **208** can be used including computer-readable storage media may

include, but is not limited to, volatile, nonvolatile, non-transitory, removable, and non-removable media implemented in any method or technology for storage of information, such as computer readable instructions, data structures, program modules, or other data. Examples of computer-readable storage media include, but are not limited to, RAM, ROM, EEPROM, flash memory, or other memory technology, CD-ROM, digital versatile disks (“DVD”) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by a processor.

(27) In one embodiment, the antenna **218** is capable of receiving signals (e.g., RF signals) from a CP/RC **206** (see, CP **18** or RC **16** of FIG. **1**) which is programmed or otherwise operated by a user. The signals sent to the processor **204** via the antenna **218** and receiver **202** can be used to modify or otherwise direct the operation of the electrical stimulation system. For example, the signals may be used to modify the pulses of the electrical stimulation system such as modifying one or more of pulse width, pulse frequency, pulse waveform, and pulse amplitude. The signals may also direct the electrical stimulation system **200** to cease operation, to start operation, to start charging the battery, or to stop charging the battery. In other embodiments, the stimulation system does not include an antenna **218** or receiver **202** and the processor **204** operates as programmed.

(28) Optionally, the electrical stimulation system **200** may include a transmitter (not shown) coupled to the processor **204** and the antenna **218** for transmitting signals back to the CP/RC **206** or another unit capable of receiving the signals. For example, the electrical stimulation system **200** may transmit signals indicating whether the electrical stimulation system **200** is operating properly or not or indicating when the battery needs to be charged or the level of charge remaining in the battery. The processor **204** may also be capable of transmitting information about the pulse characteristics so that a user or clinician can determine or verify the characteristics.

(29) Transmission of signals can occur using any suitable method, technique, or platform including, but not limited to, inductive transmission, radiofrequency transmission, Bluetooth™, Wi-Fi, cellular transmission, near field transmission, infrared transmission, or the like or any combination thereof. In addition, the IPG can be wirelessly coupled to the RC or CP using any suitable arrangement include direct transmission or transmission through a network, such as a local area network, wide area network, the Internet, or the like or any combination thereof.

(30) Returning to FIG. **1**, the CP **18** or RC **16** (or other external device) utilizes software applications to communicate with the IPG **14**. These software applications are often custom-made and instructions for the software applications can be stored in the memory of the CP, RC, or other external device. In at least some embodiments, depending on its purpose, each software application may have a capability (or permission) to perform one or more of the following: read data, write data, select from existing programs, modify programming settings of existing programs, create programs, modify or update software/firmware, full control (which could include all of the previously listed capabilities), or otherwise interact with the IPG or any combination of these capabilities (or permissions).

(31) It will also be understood that although the methods, techniques, and systems described herein are presented in the context of an external device, such as a CP or RC, communicating with an IPG, these methods, techniques, and systems can be used for communication and authentication of an external device with any medical device. In addition, the methods, techniques, and systems described herein are presented in the context of an electrical stimulation system, but it will be recognized that these methods, techniques, and systems can be used with an optical stimulation system or an electrical/optical stimulation system. Examples of optical stimulation systems or electrical/optical stimulation systems are found in U.S. Patent Application Publications Nos. 2013/0317572; 2013/0317573; 2017/0259078; 2017/0225007; 2018/0110971; 2018/0369606; 2018/0369607; 2019/0209849; 2019/0209834; 2020/0094047; and 2020/0155854 and U.S. patent application Ser. No. 16/883,404, all of which are incorporated herein by reference in their

entireties.

(32) The CP, RC, or other external device can use any suitable authentication mechanism, such as certificate or credential authentication, encrypted messaging, or both, to securely communicate with an IPG. In at least some embodiments, the authentication mechanism includes an external device (such as a CP or RC) transmitting authentication credentials, such as a digital certificate (for example, an end-entity certificate), to the IPG to authenticate the external device. The authentication credentials can be stored in the memory of the external device. Once authenticated by the IPG, the IPG authorizes access to the IPG by the external device. Examples of authentication credentials and authentication of an external device with a medical device are described in U.S. Pat. No. 9,781,086, which is incorporated herein by reference in its entirety. In at least some embodiments, the transmissions between the external device and the IPG are encrypted. Examples of encryption methods are described in U.S. Pat. No. 8,706,251, which is incorporated herein by reference in its entirety. Any other suitable encryption methods and techniques can be used including, but not limited to, public key encryption (e.g., RSA encryption) or AES encryption.

(33) A certificate or other authentication credentials are typically issued and signed by a trusted authority, such as a certificate authority. The IPG (or other medical device) can compare the certificate or other authentication credentials received from the external device with a certificate or authentication credentials (which can be stored in the memory of the IPG) that the IPG has received from the trusted authority to authenticate the external device. In at least some embodiments, the authentication credentials issued to an external device can identify one or more specific IPGs (for example, by serial number or the like) or one or more specific types of IPGs (for example, by model or serial number range or the like or any combination thereof).

(34) In at least some embodiments, the authentication and subsequent authorized access may be time-limited or session-limited. In at least some embodiments, after the time limit or session, the external device must be reauthenticated to be authorized access to the IPG. In at least some embodiments, authorized access to an IPG may also be limited by the number of connections, duration of connection, location, or the like or any combination thereof.

(35) An end-entity certificate will be used herein as an example of authentication credentials. It will be understood, however, that any other suitable authentication credentials can be substituted for the end-entity certificate and those authentication credentials can include or be modified to include the elements described herein and be used according to the methods described herein.

(36) In at least some embodiments, the end-entity certificate is organized according to the X.509 (e.g., the X.509 v3) certificate specification from the International Telecommunications Union (ITU). Any other suitable type of certificate or certificate specification can be used. The end-entity certificate can include information such as, for example, the specification and version of certificate (for example, an X.509 v3 certificate); the serial number of the certificate; the signature algorithm used to sign the certificate; the issuer name (for example, the name of the certificate authority that issued the certificate); a validity period for the certificate (for example, a start date/time and an end date/time); identifying information (for example, type, model, serial number, or the like or any combination thereof) regarding the end entity that is intended to receive the certificate (for example, the certificate subject); the public key algorithm and public key for the end entity; optional certificate extensions such as constraints, the type of certificate, certificate policies, or custom metadata that may be supplied by a manufacturer, supplier, or user of the external device or the end entity; the certificate signature algorithm; the certificate signature; or the like or any combination thereof. Other certificates can contain more or fewer elements of information.

(37) In at least some embodiments, the end-entity certificate (or other authentication credentials) that is transmitted from the external device to a medical device, such as an IPG, is not specific to a particular application on the external device. An external device, however, may have multiple applications stored thereon. For example, the external device may have different applications so that the external device can interact with different types of IPGs. In at least some embodiments,

only a subset (for example, only one or more, but less than all) of those applications are suitable for any particular type of IPG with which the external device is to communicate. For example, a CP or RC may include a first application that is suitable for use with a first type of IPG and a second application that is suitable for use with a second type of IPG. Such an arrangement can be useful so that the CP or RC is not limited to use with only a specific type or types of IPGs or other medical devices.

(38) As another example, a CP or RC may include a first application that is suitable for use by a patient and a second application that is suitable for use by a clinician. These applications may require that the individual (e.g., patient or clinician) enter a password or other identification for access to the appropriate application(s). The first and second applications may have different capabilities. For example, the patient-accessible first application may allow the patient to select from a number of different stimulation programs, but not allow the patient to modify or create stimulation programs. In contrast, the clinician-accessible second application may permit modification or creation of stimulation programs. Such an arrangement may be useful as it can provide a single device with different capabilities for different users.

(39) Because a particular CP or RC may have a number of different software applications, it can be useful to control which specific applications and devices are authorized to access specific IPGs. In at least some embodiments, application-specific end-entity certificates (or other application-specific authentication credentials) are individually (or collectively for a subset of applications) issued for one or more (or all) of the applications on an external device. In at least some embodiments, the application-specific end-entity certificate or (other application-specific authentication credentials) can securely communicate identity information about the application and the external device to the IPG in order to obtain authorized access to the IPG. For example, the end-entity certificate (or other authentication credentials) can contain information about the type(s) or identity(ies) of the IPGs that the application is authorized to communicate with or information about the application or type of application that is associated with the end-entity certificate (or other authentication credentials). In at least some embodiments, this information can be contained in the custom metadata or other optional certificate extensions of the end-entity certificate issued to that application and communicated to the IPG.

(40) In at least some embodiments, the IPG authenticates and positively identifies the external device and the application (or application type) and determines whether the external device and application (or application type) are authorized access to, and communicate with, the IPG using the information provided in the end-entity certificate (or other authentication credentials). For example, the IPG may use the information regarding the type(s) or identity(ies) of the IPG the application is authorized to communicate with or information about the application or type of application to authorize access by the external device and by the application identified in the end-entity certificate (or other authentication credentials).

(41) In at least some embodiments, the IPG terminates or prevents communication with an application or external device that is not authenticated and, therefore, not authorized to communicate and interact with that IPG. For example, if the IPG identified in the end-entity certificate (or other authentication credentials) does not match the IPG that receives the end-entity certificate (or other authentication credentials) then the IPG terminates or prevents communication with an application or external device or terminates or prevents access by the application or external device. As another example, if the application identified in the end-entity certificate (or other authentication credentials) does not match an application authorized on the IPG that receives the end-entity certificate (or other authentication credentials) then the IPG terminates or prevents communication with an application or external device or terminates or prevents access by the application or external device.

(42) FIG. 3 illustrates an IPG **314**, a first external device **350** (such as a CP or RC), and a second external device **354** (such as a CP or RC). The first external device **350** has an application with an

associated first end-entity certificate **352** (or other authentication credentials), which may be an application-specific end-entity certificate. The second external device **354** has an application with an associated second end-entity certificate **356** (or other authentication credentials), which may be an application-specific end-entity certificate.

(43) In the illustrated embodiment, the first end-entity certificate **352** identifies the IPG **314** within the first end-entity certificate **352** and, therefore, the first external device **350** is authenticated by the IPG **314** using the IPG's certificate **351** and communication **358** continues between the IPG **314** and the first external device **350** because the first external device is authorized access to the IPG. In contrast, the second end-entity certificate **356** identifies a different IPG, not IPG **314**, within the second end-entity certificate **356** and, therefore, the second external device **354** is not authenticated by the IPG **314** and communication **360** between the second external device **354** and IPG **314** or access to the IPG is terminated or not authorized.

(44) FIG. 4 illustrates an IPG **314** and an external device **350** with a first end-entity certificate **352** that is application-specific to a first application and a second end-entity certificate **356** that is application-specific to a second application. (The external device **350** is illustrated twice in FIG. 4 for ease of depiction.) In at least some embodiments, the first application-specific first end-entity certificate **352** identifies the IPG **314** and the first application within the first end-entity certificate **352** and, therefore, the application is authenticated by the IPG **314** using the IPG's certificate **351** and communication **358** continues between the IPG **314** and the first application because the first application is authorized access to the IPG. In contrast, the second application-specific second end-entity certificate **356** identifies a different IPG, not IPG **314**, within the second end-entity certificate **356** and, therefore, the second application is not authenticated by the IPG **314** and communication **360** between the second application and IPG **314** is terminated or not authorized.

(45) In other embodiments, in FIG. 4, the IPG **314** identifies the first application as authorized to interact with the IPG and, therefore, the application is authenticated by the IPG **314** using the IPG's certificate **351** and communication **358** continues between the IPG **314** and the first application because the first application is authorized access to the IPG. In contrast, the IPG **314** recognizes that the second application is not authorized to interact with the IPG and, therefore, the second application is not authenticated by the IPG **314** and communication **360** between the second application and IPG **314** is terminated or not authorized.

(46) FIG. 5 is a flowchart of one embodiment of a method of authorizing access to an IPG (or other medical device) by an external device (such as a RC or CP). In step **502**, the external device transmits authentication credentials (such as an end-entity certificate) to the IPG (or other medical device). In at least some embodiments, the authentication credentials can be application-specific. In step **504**, the IPG (or other medical device) authenticates the received authentication credentials. In step **506**, the IPG (or other medical device) determines from the received authentication credentials whether access by the external device or application of the external device is authorized. If yes, in step **508**, access by the external device or application of the external device is authorized. If no, in step **510**, communication with the external device is terminated or access by the external device or application of the external device is prevented.

(47) In at least some embodiments, an external device can include more than one application that can connect to the IPG. In at least some embodiments, the application(s) each have one or more interaction capabilities with respect to the IPG and those interaction capabilities can differ between applications. The term “permission” can be used interchangeably with the term “capability”. Examples of interaction capabilities include, but are not limited to, read data, write data, select from existing programs, modify programming settings of existing programs, create programs, modify or update software/firmware, full control (which could include all of the previously listed capabilities), or the like or any combination thereof.

(48) As one example, one application may be authorized for only read capability and another application may be authorized for full control capability. As a further example, a RC or CP could

include i) a patient application that has, for example, read and program selection capabilities; ii) a clinician application which has, for example, read, write, program selection, modifying programming settings of existing programs, and creating programs (or even has full control); and iii) a manufacturer application that has, for example, full control or read, write, and software/firmware update capabilities. Other applications, including other patient, clinician, or manufacturer applications, can have different interaction capabilities or sets of interaction capabilities.

(49) Each application is limited to only the interaction capabilities authorized for that application. Examples of other interaction capabilities or interaction capability restrictions can include, for example, security-based limitations, clinician-controlled settings, and remote therapy adjustment.

(50) In at least some embodiments, an application-specific end-entity certificate (or other application-specific authentication credentials) can include interaction capability information about the application. For example, information about the application's interaction capabilities can be contained in the custom metadata or other certificate extensions of the end-entity certificate issued to that application and communicated to the IPG. In at least some embodiments, the interaction capabilities in the application-specific end-entity certificate (or other application-specific authentication credentials) include all of the interaction capabilities that the application is capable of. In at least some embodiments, the interaction capabilities may be determined by (or in view of) or limited by (or in view of) the signing entity (e.g., certificate authority or a sub-certificate authority) of the end-entity certificate (or other authentication credentials).

(51) In other embodiments, the interaction capabilities in the application-specific end-entity certificate (or other application-specific authentication credentials) may not include all of the interaction capabilities that the application is capable of, but only a subset of those capabilities. For example, an application may be useable by both a patient and a clinician, but the application on the patient's RC may have an end-entity certificate (or other application-specific authentication credentials) that only includes, for example, read and program selection capabilities. The application on a clinician's CP may have an end-entity certificate (or other application-specific authentication credentials) that includes, for example, read, write, program selection, modifying programming settings of existing programs, and creating programs (or even includes full control).

(52) Alternatively, in at least some embodiments, an end-entity certificate (or other authentication credentials) for an external device may specify interaction capabilities for all applications of that external device. For example, an RC may have an end-entity certificate (or other application-specific authentication credentials) that only has, for example, read and program selection capabilities. A CP may have an end-entity certificate (or other application-specific authentication credentials) that has, for example, read, write, program selection, modifying programming settings of existing programs, and creating programs (or even has full control).

(53) In at least some embodiments, the IPG authenticates or validates the application capabilities using the end-entity certificate (or other authentication credentials). In at least some embodiments, the IPG authenticates or validates the application's interaction capabilities by identifying the capabilities listed within the custom metadata or other certificate extensions of an application-specific end entity certificate.

(54) In at least some embodiments, the IPG only allows or authorizes access to the IPG based on the interaction capabilities listed for the application or external device in the end entity certificate (or other credentials). For example, if the application's interaction capabilities in the end-entity certificate (or other authentication credentials) only allow read access, the IPG prevents write operations.

(55) FIG. 6 illustrates an IPG **614**, a first external device **650** (such as a RC), a second external device **654** (such as a CP), and a third external device **662** (such as a second RC). The first external device **650** has an application with an associated application-specific first end-entity certificate **652** (or other authentication credentials). The second external device **654** has a second end-entity

certificate **656** (or other authentication credentials) for the second external device. The third external device **662** has a third end-entity certificate **652** (or other authentication credentials) for the third external device.

(56) In the illustrated embodiment, the application-specific first end-entity certificate **652** identifies the IPG **614** within the first end-entity certificate **652** and indicates that the application has only read capability. Therefore, the application of the first external device **650** is authenticated by the IPG **614** using the IPG's certificate **651** and communication **658** continues between the IPG **614** and the application of the first external device **650** because the application of the first external device is authorized access to the IPG. The application of the first external device **650**, however, has only the read capability.

(57) The second end-entity certificate **656** identifies the IPG **614** within the second end-entity certificate **656** and indicates that the second external device **654** has full control capability. Therefore, the second external device **654** is authenticated by the IPG **614** using the IPG's certificate **651** and communication **660** continues between the IPG **614** and the second external device **654** because the second external device is authorized access to the IPG. The applications of the second external device **654** have full control capability.

(58) The third end-entity certificate **664** identifies the IPG **614** within the second end-entity certificate **664** and indicates that the third external device **664** has read, write, modify programming settings of existing programs, and create programs capabilities. Therefore, the third external device **664** is authenticated by the IPG **614** using the IPG's certificate **651** and communication **666** continues between the IPG **614** and the third external device **664** because the third external device is authorized access to the IPG. The applications of the third external device **654** have read, write, modify programming settings of existing programs, and create programs capabilities.

(59) FIG. 7 is a flowchart of one embodiment of a method of authorizing access to an IPG (or other medical device) by an external device. In step **702**, the external device transmits authentication credentials (such as an end-entity certificate) to the IPG (or other medical device). The authentication credentials include an indication of one or more interaction capabilities of the external device or one or more applications of the external device. In at least some embodiments, the authentication credentials can be application-specific. In step **704**, the IPG (or other medical device) authenticates the received authentication credentials. In step **706**, the IPG (or other medical device) determines from the received authentication credentials whether access by the external device or application of the external device is authorized. If yes, in step **708**, access by the external device or application of the external device is authorized. If no, in step **710**, communication with the external device is terminated or access by the external device or application of the external device is prevented.

(60) In step **712**, there is a determination whether the external device (or an application of the external device) is interacting with the IPG. If so, in step **714**, the IPG determines whether the interaction falls within the interaction capabilities specified in the authentication credentials. If so, in step **716**, the IPG and external device proceed with the interaction. If not, in step **718**, the IPG prevents the interaction.

(61) In some embodiments, an authorized first external device can facilitate the authentication or authorization of a second external device. For example, a RC that is authorized, using authentication credentials as described above, to access an IPG can facilitate the authentication or authorization of a CP to provide authorized access to the IPG by the CP. This can be particularly useful when, for example, the RC (which is intended to communicate with a specific IPG) has authorization credentials that specify the IPG but the CP (which is intended to communicate with any suitable IPG) does not.

(62) In at least some embodiments, the first external device can present authentication credentials to the IPG (or other medical device) and thereby obtain authorized access to the IPG, as described above. The second external service presents authentication credentials to the first external device

which confirms the authentication credentials. For example, the first external device can compare the authentication credentials from the second external device to the authentication credentials (e.g., a certificate from a trusted authority) on the first external device. After verification, the first external device provides a token or other authorization credentials to either the IPG or the second external device that authorizes access to the IPG by the second external device.

(63) FIG. 8 illustrates an IPG **814**, a first external device **850** (such as a RC), and a second external device **854** (such as a CP). The first external device **850** has first authentication credentials **852** (such as a first end-entity certificate), which may be application-specific authentication credentials. The second external device **854** has second authentication credentials **856**, which may be application-specific authentication credentials.

(64) In the illustrated embodiment, the first authentication credentials **852** identifies the IPG **814** within the first authentication credentials **852** and, therefore, the first external device **850** is authenticated by the IPG **814** using the IPG's certificate **851** and communication **858** continues between the IPG **814** and the first external device **850** because the first external device is authorized access to the IPG.

(65) The second external device **854** does not have authentication credentials that specify IPG **314** and so the second external device cannot directly authenticate with the IPG. The second external device **854** communicates **860** with the first external device **850** and presents the second authentication credentials **856** to the first external device. The first external device **850** authenticates the second external device **854** using the second authentication credentials **856** and authorizes access to the IPG **314** which may be based on the authorized access that the first external device has obtained using its own first authentication credentials **852**. The first external device **850** provides either the second external device **854** or the IPG **314** with a token or other authorization credentials so that the second external device can communicate **862** with, and have authorized access to, the IPG **314**.

(66) In some embodiments, interaction capabilities, as described above, may be specified in one or both of the first authentication credentials **852** or the second authentication credentials **856**. In at least some embodiments, the first external device **850** can authorize any interaction capabilities specified in the second authentication credentials **856**. In at least some embodiments, the first external device **850** can authorize only those interaction capabilities specified in the first authentication credentials **852**.

(67) FIG. 9 is a flowchart of one embodiment of a method of authorizing access to an IPG (or other medical device) using a first external device to authorize access by a second external device. In step **902**, the first external device transmits authentication credentials (such as an end-entity certificate) to the IPG (or other medical device). In at least some embodiments, the authentication credentials can be application-specific. In step **904**, the IPG (or other medical device) authenticates the received authentication credentials to authorize access by the first external device (see, for example, the methods described above). In step **906**, the second external device transmits authentication credential to the first external device. In step **908**, the first external device authenticates the received authentication credentials from the second external device to authorize access to the IPG by the second external device. In step **910**, the first external device provides a token or other authorization credentials to the second external device or IPG or both. In step **912**, the IPG and second external device use the token or other authorization credentials for authorized access to the IPG by the second external device.

(68) In at least some embodiments, the token or other authorization credentials can be limited by time or limited to a session or by any other suitable limitation(s). Such limitation(s) may be imposed by the first external device, IPG, or second external device or any combination thereof.

(69) Any of the systems or methods described herein may also incorporate two-factor authentication. For example, the systems or methods may also require that the user provide identification to the external device, such as a password, fingerprint, or code that has been sent to

another user device, or the like or any combination thereof. In at least some embodiments, the IPG may require two-factor authentication by entry of a password or by application of a magnet near the IPG. For example, a specified sequence of on-off steps may be required by the IPG for the second authentication factor where “on” indicates that the magnet is near the IPG and “off” indicates that the magnet has been removed from proximity to the IPG. One example of such a sequence can be “on”-“off”-“on”.

(70) In at least some embodiments, a patient may provide a fingerprint, password, or other identification to register a first external device (such as a RC) to the patient. The first external device can provide the fingerprint, password, or other identification to the IPG. Subsequently, each second external device that is to be authenticated with the IPG will also provide the fingerprint, password, or other identification along with other authentication credentials to demonstrate that the patient has given consent to access to the IPG. In at least some embodiments, the patient may request that the first external device provide the fingerprint, password, or other identification to the second external device. Such a request may require that the patient provide or enter the fingerprint, password, or other identification into the first external device.

(71) It will be understood that each block of the flowchart illustration, and combinations of blocks in the flowchart illustration and methods disclosed herein, can be implemented by computer program instructions. These program instructions may be provided to a processor to produce a machine or engine, such that the instructions, which execute on the processor, create means for implementing the actions specified in the flowchart block or blocks or engine disclosed herein. The computer program instructions may be executed by a processor to cause a series of operational steps to be performed by the processor to produce a computer implemented process. The computer program instructions may also cause at least some of the operational steps to be performed in parallel. Moreover, some of the steps may also be performed across more than one processor, such as might arise in a multi-processor computing device. In addition, one or more processes may also be performed concurrently with other processes, or even in a different sequence than illustrated without departing from the scope or spirit of the invention.

(72) The computer program instructions can be stored on any suitable computer-readable medium including, but not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (“DVD”) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by a computing device. The computer program instructions can be stored locally or nonlocally (for example, in the Cloud).

(73) The above specification and examples provide a description of the arrangement and use of the invention. Since many embodiments of the invention can be made without departing from the spirit and scope of the invention, the invention also resides in the claims hereinafter appended.

Claims

1. An external device for wireless communication with an implantable medical device, the external device comprising: a memory comprising first authentication credentials stored thereon; and a processor coupled to the memory and configured to perform actions comprising: transmitting the first authentication credentials directly to the implantable medical device for the implantable medical device to authorize access to the implantable medical device by the external device, wherein the first authentication credentials comprise a first certificate issued to the external device by a certificate authority, wherein the first certificate identifies the external device; receiving a request to facilitate authorization of a second device by the implantable medical device and receiving second authentication credentials from the second device, wherein the second authentication credentials comprise a second certificate issued to the second device by a certificate authority, wherein the second certificate identifies the second device; determining authentication or

authorization of the second device using the second authentication credentials; and after authorization of the external device by the implantable medical device and determining authentication or authorization of the second device, transmitting a token or third authorization credentials to at least one of the implantable medical device or the second device directing the implantable medical device to authorize direct access to the implantable medical device by the second device so that both the external device and the second device have authorized direct access to the implantable medical device.

2. The external device of claim 1, wherein the memory has further stored thereon instructions for a first application configured to interact with implantable medical devices, wherein the first certificate comprises explicit identification of at least one interaction capability of the first application with respect to the implantable medical device, wherein each of the at least one interaction capability identifies a specific interaction with the implantable medical device, wherein at least one of the at least one interaction capability is selected from i) selecting from existing stimulation programs, ii) modifying programming settings of existing stimulation programs, or iii) creating stimulation programs; wherein the actions further comprise: executing the first application using the instructions; and transmitting the first authentication credentials to the implantable medical device for the implantable medical device to authorize access to the implantable medical device by the first application limiting interaction between the first application and the implantable medical device to the at least one interaction capability explicitly identified in the first certificate.

3. The external device of claim 2, wherein the memory further has stored thereon instructions for a second application configured to interact with implantable medical devices and additional authentication credentials specific to the second application.

4. The external device of claim 2, wherein the first authentication credentials comprise identification of at least one specific implantable medical device or at least one specific type of implantable medical device that the first application is configured to interact with.

5. The external device of claim 2, wherein the first certificate comprises an end-entity certificate.

6. The external device of claim 5, wherein the end-entity certificate comprises an identification of the first application in custom metadata or certificate extensions of the end-entity certificate.

7. The external device of claim 2, wherein the at least one interaction capability is at least two interaction capabilities, wherein at least another one of the at least two interaction capabilities is selected from reading data, writing data, or modifying or updating software/firmware.

8. A system, comprising: the external device of claim 2; and the implantable medical device configured for wireless communication with the external device.

9. The system of claim 8, wherein the first authentication credentials comprise identification of the implantable medical device in the first authentication credentials.

10. The system of claim 8, wherein the implantable medical device is configured to limit authorized access to the implantable medical device by the first application to only the at least one interaction capability identified in the first authentication credentials.

11. The external device of claim 1, wherein the token or third authorization credentials are limited to a specified time period.

12. The external device of claim 1 wherein the token or third authorization credentials are limited to a single session.

13. A system, comprising: the external device of claim 1; and the second device comprising a memory having stored thereon instructions for a second application configured to interact with implantable medical devices and the second authentication credentials, and a processor coupled to the memory and configured to perform actions comprising: executing the second application using the instructions; and transmitting the second authentication credentials to the external device.

14. The system of claim 13, wherein the memory of the external device has further stored thereon instructions for a first application configured to interact with implantable medical devices, wherein the first authentication credentials are specific to the first application and identify at least one

interaction capability of the first application with respect to the implantable medical device wherein the second authentication credentials identify of at least one interaction capability of the second application with respect to the implantable medical device.

15. The system of claim 14, wherein the token or third authorization credentials identify the at least one interaction capability of the first application.

16. The system of claim 14, wherein the token or third authorization credentials identify the at least one interaction capability of the second application.

17. The system of claim 13, further comprising the implantable medical device configured for wireless communication with the external device and the second device.

18. The system of claim 17, wherein the implantable medical device is configured to limit authorized access to the second device based on identification of at least one interaction capability in the token or third authorization credentials.

19. The system of claim 1, wherein transmitting the token or third authorization credentials comprises transmitting the token or third authorization credentials to the implantable medical device directing the implantable medical device to authorize direct access to the implantable medical device by the second device.

20. A system, comprising: the external device of claim 1; and the implantable medical device configured for wireless communication with the external device.
